Research Article

Towards NAM-Based Risk Assessment for Developmental Neurotoxic Effects, Illustrated with Chlorpyrifos

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Abstract

New approach methodologies (NAMs), such as *in vitro* and *in silico* methods, provide opportunities to replace, reduce, and refine animal studies for chemical risk assessment, decrease experimental lead time, and increase predictivity for human health. However, starting from or including NAMs introduces uncertainties that need to be addressed and quantified to work towards regulatory implementation of NAMs within risk assessment. A theoretical risk assessment based on the developmental neurotoxicity (DNT) of chlorpyrifos and its active metabolite chlorpyrifos-oxon was performed to highlight important extrapolations and identify and quantify uncertainties when extrapolating NAM-based data to estimate human health risk. Cell viability of differentiating human neuronal progenitor cells was used as a proxy for DNT during early pregnancy. A physiologically based kinetic (PBK) model describing early pregnancy was developed to facilitate quantitative *in-vitro*-to-*in-vivo* extrapolation (qIVIVE) to estimate the external dose leading to DNT onset. An *in vitro*-based health-based threshold value (HBTV) was derived for the onset of DNT caused by chlorpyrifos and chlorpyrifos-oxon following chlorpyrifos exposure. This value was higher than the maternal exposure associated with the onset of DNT based on epidemiological data, showing the need for improvements in the NAM-based risk assessment of chemicals suspected of DNT. Improvements relate to the selection and refinement of the *in vitro* endpoint and assay and defining the quantitative relationship between the selected key event and the adverse outcome and provide a starting point for future research to aid regulatory implementation.

Plain language summary

Using new approach methodologies (NAMs) in chemical risk assessment, like cell tests and computer models, can help to replace, reduce, and refine animal studies and improve the predictivity of the results for human health. We used NAMs to estimate the daily amount of the pesticide chlorpyrifos that can be ingested during pregnancy without affecting the development of the fetal nervous system. Considering the differences between the experimental measurements and real-life, we translated the measurements and calculations to a predicted safe daily dose in humans. Additionally, uncertainties were identified and, when possible, quantified. However, the resulting safe daily dose was higher than exposures measured in humans where neurodevelopmental effects had been observed. We need to further improve our understanding of the use of NAMs in the risk assessment of suspected neurodevelopmental toxicants.

1 Introduction

Traditionally, the toxicological risk assessment of chemicals has been based on animal studies. However, animal studies can be time and resource intensive and are increasingly considered undesirable because of ethical considerations (Hartung, 2008, 2017). Moreover, their predictivity for human health may be limited due to interspecies differences (Hartung, 2008, 2017). Nowadays, many new approach methodologies (NAMs), e.g., *in silico* and *in vitro* approaches, are being developed to replace, reduce, and refine animal studies (3Rs). The use of NAMs in risk assessment provides many opportunities, including determining the internal or external exposure below which no adverse effects in humans are expected to occur, while aligning with the 3Rs by using test systems that are more relevant to humans (Dent et al., 2018).

In vitro human cell assays can provide insight into molecular and cellular effects and can identify possible hazards of compounds. *In vitro* data can be extrapolated to human equivalent doses (HEDs) by means of quantitative *in-vitro*-to-*in-vivo* extrapolation (qIVIVE). Whereas the use of human cells eliminates the uncertainties related to interspecies differences in sensitivity to a chemical, other extrapolations are needed that introduce additional uncertainties.

The importance of (quantitatively) addressing these uncertainties and extrapolations was recently highlighted by Zwartsen et al. (2025) using a case study on liver steatosis for the fungicide imazalil on the Monte Carlo Risk Assessment (MCRA) platform. Figure 1 provides an overview of the steps that should ideally be performed for hazard characterization using *in vitro* data, based on Zwartsen et al. (2025). These steps cover the extrapolations inherent to qIVIVE, i.e., the extrapolation of the molecular perturbation measured *in vitro* (e.g., a key event in an adverse outcome pathway (AOP)) to an adverse health outcome in humans (A), the extrapolation of nominal to free or intracellular *in vitro* concentrations (B), the extrapolation of internal concentrations to external doses in humans (C), and if animal cells are used, the extrapolation between species (D). To obtain an *in vitro*-based health-based guidance value (HBGV) from an *in vitro*-based HED, additional extrapolations related to differences in *in vitro* and *in vivo* exposure durations (E) and interindividual differences (F) should be addressed. If a different endpoint than the most sensitive endpoint of a certain chemical is studied, an *in vitro*-based health-based threshold value (HBTV) can be derived instead of an *in vitro*-based HBGV. Reasoning for looking at endpoints other than the most sensitive endpoint is hazard screening. Additionally, determining a HBTV instead of a HBGV aids, e.g., 1) comparing values to chemicals with similar effect, 2) cumulative risk assessment for chemicals that are part of an assessment group, and 3) comparing with epidemiological data to see if the seen effect could be induced by a certain chemical.

Developmental neurotoxicity (DNT), i.e., adverse changes in the developing brain, is a toxicological endpoint of concern for many compounds (OECD, 2023). One chemical class suspected of DNT effects is the group of anticholinesterase insecticides (Wołejko et al., 2022; OECD, 2023). These chemicals were developed for their ability to inhibit acetylcholinesterase (AChE) in the nervous system of insects, which leads to nerve poisoning and death (Karise and Mänd, 2015). Since AChE is highly conserved across species, these insecticides also affect other species through the same molecular mechanism (Richardson, 2018). Unfortunately, not much is known about how AChE inhibition could lead to DNT in humans. According to Slikker (2018), the mechanisms underlying about two-thirds of the known anatomical and functional developmental changes induced by suspected DNT-inducing compounds have not yet been identified. Additionally, there is growing evidence that anti-AChE compounds can also negatively affect other non-AChE targets, and these mechanisms might be leading to DNT as well (Carr et al., 2018; OECD, 2023).

Current regulatory approaches to determine the potential DNT of chemicals in humans are based on *in vivo* test protocols that are resource intensive in terms of cost, time, and number of animals used. Recently, efforts were made to develop NAMs for DNT testing, including the development and review of a DNT *in vitro* testing battery (OECD, 2023). The Organisation for Economic Co-operation and Development (OECD) recently published an initial recommendation on the evaluation of data from this DNT *in vitro* testing battery (OECD, 2023). They identified gaps for qIVIVE, including the need to incorporate placental transfer, the possibility to estimate fetal and postnatal brain concentrations, and to include postnatal development of the blood-brain-barrier in used models. Additionally, the OECD highlighted that there is currently a lack of internationally accepted guidance on IVIVE procedures and models.

Recently, Algharably et al. (2023) covered some of these gaps (i.e., calculations related to placental transfer and fetal brain concentrations) by applying physiologically based kinetic (PBK) modelling and reverse dosimetry to derive a benchmark dose (BMD) for DNT following maternal exposure to the anti-AChE insecticide chlorpyrifos (CPF) based on *in vitro* DNT data from Di Consiglio et al. (2020). DNT effects of CPF have been observed in *in vivo* animal studies and in epidemiological studies in children with prenatal exposure to CPF (EFSA, 2019a; Rauh et al., 2006, 2011, 2012; Chiu et al., 2021). Reported neurodevelopmental effects in infants following maternal exposure to CPF include motor function deficit (Silver et al., 2017), worse cognitive and language performance (Chiu et al., 2021), lower psychomotor and mental development scores, increased attention deficit/hyperactivity disorder problems (Rauh et al., 2006), and lower IQ and reduced working memory (Rauh et al., 2011).

Based on DNT effects as a result of AChE inhibition, the European Food Safety Authority (EFSA) identified a lowest-observed-adverse-effect-level (LOAEL) of 0.3 mg/kg bw from a study in rats as point of departure (PoD) (EFSA, 2019a). However, no reference values (HBGV) could be determined due to uncertainties about genotoxicity and the DNT study. In the rat DNT study, effects were observed at all doses tested (EFSA, 2019a). EFSA and the United States Environmental Protection Agency (US EPA) raised additional concerns related to DNT: Chemicals such as CPF might also have activities other than AChE inhibition that lead to DNT (EFSA, 2019a; US EPA, 2020).

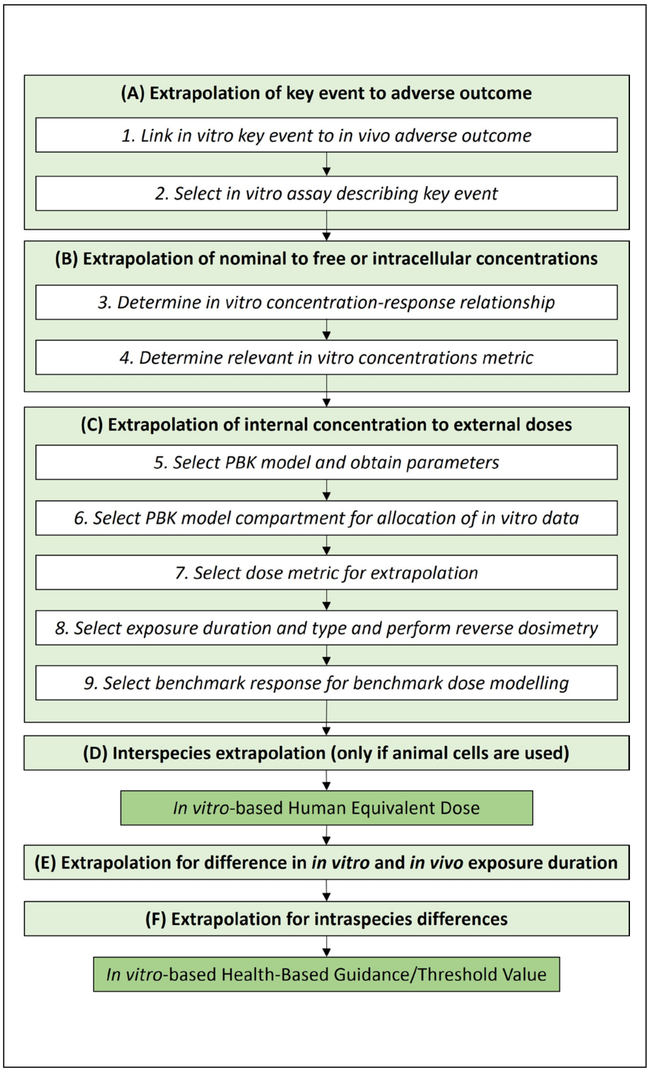

Fig. 1: Overview of the qIVIVE steps (white boxes, italics) and extrapolations (light green boxes, bold) that should be accounted for when calculating an *in vitro*-based human equivalent dose (*in vitro*-based HED), and subsequently an *in vitro*-based health-based guidance value (*in vitro*-based HBGV – in case of the critical effect) or an *in vitro*-based health-based threshold value (*in vitro*-based HBTV) (dark green boxes)
This figure is based on the steps and extrapolations listed in Zwartsen et al. (2025).

Using NAMs, Algharably et al. (2023) predicted an HED based on *in vitro* non-cholinergic toxicodynamic data and reported that it was several orders of magnitude higher than the PoD from the rat study. Additionally, they calculated HEDs using epidemiological data, which were several orders of magnitude lower than the PoD from the rat study. This discrepancy could possibly be explained by a lack of sensitivity of the used *in vitro* cell model, by a lack of consideration of active metabolites in the study, and/or by the PoD of the rat study. As seen in this study, performing a risk assessment based on NAMs requires further development and additional case studies.

The aim of the current study was to perform an illustrative NAM-based risk assessment for the induction of DNT, independent of cholinesterase inhibition, by CPF and its primary metabolite chlorpyrifos-oxon (CPO), while addressing the steps and extrapolations required for qIVIVE described by Zwartsen et al. (2025). Although the use of CPF has been banned in the EU (Regulation EU 2020/18 (EC, 2020)), we consider CPF a good model compound, since it is relatively data-rich, which allows us to illustrate the hazard characterization steps needed to perform a NAM-based risk assessment for a DNT-inducing chemical. CPF and its primary metabolite CPO are both known to induce neurotoxic effects *in vitro* (Howard et al., 2005; Monnet-Tschudi et al., 2000). Observed neurotoxic effects include cytotoxicity of neuronal and glial cells and a decrease in total axonal length, which are both linked to DNT (Howard et al., 2005; Monnet-Tschudi et al., 2000). These effects were not observed for 3,5,6-trichloro-2-pyridinol (TCPy), the metabolite of CPF and CPO (Howard et al., 2005). The case study was performed using an in-house *in vitro* assay with differentiating human neural progenitor cells (NPCs) and a newly developed PBK model, both mimicking and describing early pregnancy.

2 Materials and methods

This section follows the 9 steps of Figure 1 to derive an *in vitro*-based HED and consequently obtain an *in vitro*-based HBTV.

2.1 Hazard characterization steps

2.1.1 Step 1: Link *in vitro* key event to *in vivo* adverse outcome, and Step 2: Select *in vitro* assay describing key event

Currently, a quantitative understanding between the activation of key events *in vitro* and the activation of the adverse outcome DNT *in vivo* is lacking. Based on a network of qualitative AOPs for (developmental) neurotoxicity (Spinu et al., 2019), “Increase, Cell injury/death” was selected as the key event of interest, since it is the key event included in most AOPs of the AOP network. Even though a more specific *in vitro* endpoint or a battery of *in vitro* tests may be preferred, quantitatively relating any *in vitro* assay to an *in vivo* adverse outcome seen as DNT is challenging. The extrapolation of the *in vitro* key event to the *in vivo* adverse outcome is addressed in Section 4 (Extrapolation A).

Cell death can be approached from a variety of viewpoints, including quantification of live cells (viability). We determined the viability of differentiating NPCs using the WST-1 assay. The WST-1 assay relies on the ability of viable cells to convert a tetrazolium salt to a colored formazan product, which is associated with metabolic activity and serves as a proxy for cell viability (Méry et al., 2017). The human neural progenitor test (hNPT) was used since the differentiating NPCs in this test mimic brain development during early pregnancy and show neuronal functionality relatively quickly compared to other neural differentiation assays (de Leeuw et al., 2020, 2022).

2.1.2 Step 3: Determine the concentration-response relationship

Compounds
Chlorpyrifos (CPF, CAS# 2921-88-2, ≥ 98% purity) was purchased from Sigma-Aldrich (St. Louis, MO, USA) and chlorpyrifos-oxon (CPO, CAS# 5598-15-2, ≥ 97% purity) from Dr. Ehrenstorfer (Augsburg, Germany). Stock solutions were freshly prepared for each experiment in dimethyl sulfoxide (DMSO, CAS# 67-68-5, Sigma-Aldrich). Final solutions contained a concentration of 0.25% DMSO (v/v). Stock solutions and final solutions were prepared in polypropylene vials.

The human neural progenitor test (hNPT) differentiation protocol
H9 human embryonic stem cells (WA09, passage 26, WiCell, Madison, WI, USA) were used to generate neural progenitor cells (NPCs) as described previously (de Leeuw et al., 2020). In short, NPCs were cultured in STEMdiff™ Neural Progenitor Medium (Stemcell, Vancouver, Canada) on poly-L-ornithine (PLO, 15 µg/mL, Sigma-Aldrich, Saint Louis, MO, USA)-laminin (10 µg/mL, Sigma-Aldrich) coated 6-well plates (Corning, New York, NY, USA) in a humidified chamber (37°C, 5% CO_2, 3% O_2). Culture medium was refreshed daily for seven days until the NPCs reached 100% confluency.

To initiate neuronal-astroglial differentiation, NPCs were dissociated and cultured at 2.56×10^5 cells/cm^2 on PLO-laminin-coated 96-well plates (Greiner Bio-One, Kremsmünster, Austria). Cells were grown in STEMdiff™ Neural Progenitor Medium for one day, which was then replaced by neural differentiation medium, marking the start of the hNPT. Neural differentiation medium was adapted from Gunhanlar et al. (2017) and is comprised of neurobasal medium, 10 µL/mL N2 supplement (100x), 20 µL/mL B-27 without retinoic acid supplement (50x), 10 µL/mL 5000 IU/mL penicillin/5000 µg/mL streptomycin, 10 µL/mL nonessential amino acids (100x), 20 ng/mL recombinant glial cell line-derived neurotrophic factor (GDNF), 20 ng/mL recombinant brain-derived neurotrophic factor (BDNF),10 ng/mL recombinant ciliary neurotrophic factor (CNTF; all from Gibco, Waltham, MA, USA), 1 µM dibutyryl cyclic adenosine monophosphate (Sigma-Aldrich), 200 µM ascorbic acid (Sigma-Aldrich), and 2 µg/mL laminin (Sigma-Aldrich)). GDNF, BDNF, CNTF, ascorbic acid, and dibutyryl cyclic adenosine monophosphate were all dissolved in demineralized water and stored in sterile solutions at -20°C until use. Cells received half medium refreshments on days 3 and 7 and were cultured for up to 10 days.

Inter-individual differences (Extrapolation F) in toxicodynamics were not measured *in vitro*, since solely one donor (of H9 cells) was used to generate NPCs. Differences in toxicodynamics could be accounted for by using various donors. Alternatively, we accounted for toxicodynamic differences, combined with toxicokinetic differences, as specified in Section 2.1.9.

Characterization of in vitro *model*
The tool CoNTExT was used to benchmark the hNPT developmental stage against human brain development (Stein et al., 2014), specifically the rank-rank hypergeometric overlap (RRHO) mapping tool of which the code is available on GitHub[1]. Whole transcriptome data (35,076 non-zero transcripts) of the NPCs and hNPT at day 10 was used as input data. This data is available upon request.

In vitro *exposure and cell viability measurement*
In the hNPT, cells were exposed to 0.001, 0.0033, 0.01, 0.033, 0.1, 0.33 or 1.0 mM CPF or CPO for ten days from the moment they were switched to differentiation medium. For the half medium refreshments on days 3 and 7, the same final concentration of compounds was added as on day 0. Three independent experiments with 6 technical replicates per condition were performed. Following the first experiment, the lowest and highest dose were excluded from one experiment. For cell viability measurements at day 10, cells were incubated for one hour with Cell Proliferation Reagent WST-1 (Roche, Mannheim, Germany) (de Leeuw et al., 2022). Extinction values were measured at 544Ex/590Em nm on a SpectraMax® M2 spectrofluorometer (Molecular Devices, Berkshire, United Kingdom).

2.1.3 Step 4: Determine relevant *in vitro* concentration metric and extrapolate nominal to free or intracellular concentrations (Extrapolation B)

We assume that the inhibition of the metabolic activity of differentiating NPCs is directly related to intracellular concentrations of CPF and CPO. *In situ* formation of CPO is considered unlikely, given the low gene expression of the relevant metabolizing enzymes (i.e., CYP1A2, CYP2B6, CYP2C19 and CYP3A4; NCBI GEO database, accession number GSE166297). The intracellular CPF and CPO concentrations to which the cells were exposed were calculated based on the nominal concentrations using the method from Kramer (2010). The calculation of the distribution of CPF and CPO in the *in vitro* system is based on the lipophilicity of the compounds (Annex 2, Tab. S2[2]), type of cells, type of medium, and type of well plate used (Section 2.1.2). The cellular volume of NPCs and the fractions of water, lipids, and proteins in NPCs were assumed to be equal to those of SH-SY5Y cells (Lee et al., 2021). Since the exposure medium was refreshed during the exposure duration, *in vitro* distribution was also calculated following medium refreshment. Nominal, free in medium, and intracellular amounts were used to determine the concentration-response relationships (Step 3). The intracellular concentration was calculated by dividing the cell-associated amount by cellular

[1] https://github.com/dhglab/RRHO_RNAseq
[2] doi:10.14573/altex.2412051s

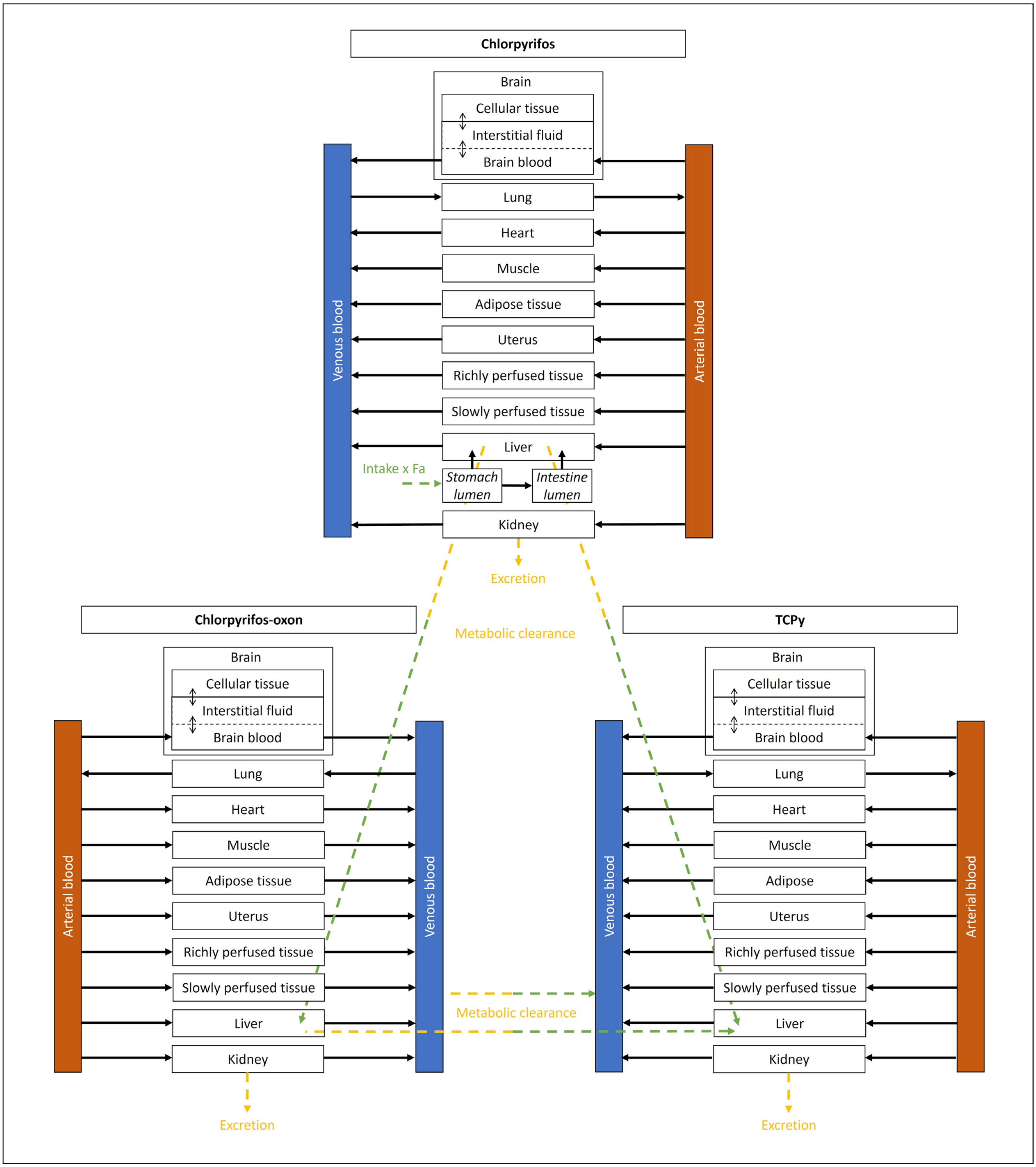

Fig. 2: Schematic diagram of the PBK model for CPF and its metabolites CPO and TCPy
The influx (oral uptake or formed metabolite in liver or blood) and efflux (urinary excretion and metabolic clearance) routes of CPF and its metabolites are visualized with green and yellow arrows, respectively.

volume. The concentration-response curves were visualized using PROAST software (version 70.3) in R (version 4.2.0)[3] (see [4] and RIVM, 2021).

2.1.4 Step 5: Select PBK model and obtain parameters

PBK model development

The PBK model for CPF and its two major metabolites, CPO and TCPy, was developed based on the models from Timchalk et al. (2002) and Zhao et al. (2022), with some modifications. CPO is included since this metabolite is known to affect biological mechanisms related to DNT, including AChE inhibition (Monnet-Tschudi et al., 2000), and TCPy is the main metabolite detected in urine. As the *in vitro* assay mimics several cellular processes during early pregnancy (8-16 weeks of gestation), which relates to the differentiation of human neuronal progenitor cells to neurons and astrocytes (de Leeuw et al., 2022), we aimed to represent a female adult at reproductive age (18-45 year) in early pregnancy in our PBK model. Because at this stage of pregnancy placental or other barriers (Tetro et al., 2018; Turco and Moffett, 2019) and fetal organs have not been fully formed, we assumed that the concentrations in the fetus are equal to the concentrations in the uterus. Therefore, we only added a uterus compartment and no placenta or specific fetal compartment(s). The inclusion of a sub-model for the fetus was considered but not chosen since it would introduce many uncertainties related to the complexity of parametrization and mathematical description of the fetus during this time frame. With regard to the physiological changes in body composition of the mother, we assumed these to be minimal during early pregnancy, and these were therefore not taken into account.

For future applications of the PBK model, target tissues often investigated using *in vitro* assays (e.g., the brain) are included in the model. The model for CPF and the submodels for CPO and TCPy all contain ten compartments for the brain, lungs, heart, muscle, adipose tissue, uterus, liver, kidneys, and remaining richly and slowly perfused tissues. Oral exposure to CPF was modeled by gastro-intestinal tract uptake. The absorption of CPF from the gastro-intestinal tract to the liver was described using a stomach lumen compartment and an intestinal lumen compartment to account for lag time, modelled independent of blood flow (for more information see Timchalk et al., 2002), and was dependent on the fraction absorbed (Fa) and absorption rates (see Section 2.1.4, *Obtaining model parameters*). The stomach and intestinal organ tissues themselves are included in the richly perfused tissue compartments in all three submodels. Following uptake, CPF is distributed throughout the body. CPF is metabolized in the liver to CPO and TCPy. CPO can also be metabolized to TCPy in both the liver and in blood. Based on human data, we assumed that CPF is only excreted in the form of TCPy via urine (Nolan et al., 1984). As a result, urinary excretion constants of CPF and CPO were set to zero.

The brain is described as a permeability-limited compartment due to the presence of the blood-brain barrier, which affects the influx and efflux of substances (Kasteel et al., 2021). The transfer of substances between the brain sub-compartments (brain tissue and brain blood) is described by a partition coefficient and a permeability surface area constant (see Annex 2[2] for more information). All other tissue compartments are described as perfusion-limited compartments and are connected by arterial and venous blood flows. For CPO and TCPy, we decided to apply the same model structure as for CPF, but we removed the stomach and intestinal lumens.

Figure 2 presents the final structure of the PBK model, which represents a female adult at reproductive age (18-45 years) and describes the absorption, distribution, metabolism and excretion (ADME) of CPF and its metabolites CPO and TCPy.

The PBK model for CPF, CPO, and TCPy was coded using Berkeley Madonna software (Version 8.0.1, Macey and Oster, UC Berkeley, California). Rosenbrock's algorithm for stiff systems was used to run the differential equations and mass balance. The model code is provided in Annex 3[2].

Obtaining model parameters

The values of all physicochemical parameters and chemical-specific parameters for each substance can be found in Annex 2, Table S1 and S2[2]. It was not possible to obtain parameters solely describing female ADME, so available (often mixed gender) values were used. The relevant physiological parameters of females at reproductive age (e.g., body weight, volumes of different organs, blood flows, etc.) used in the PBK model are listed in Annex 2, Table S3[2].

Parameters for which assumptions were made or that were fitted are described here. Absorption rate constants K_{aS} and K_{aI} describe the absorption rates from the stomach lumen to the liver and from the intestinal lumen to the liver, respectively. K_{sI} describes the transfer rate constant from the stomach lumen to the intestinal lumen. These constants were obtained using the parameter curve fitting function of Berkeley Madonna (Version 8.0.1). The values were fitted using the human *in vivo* TCPy concentration-time curves in blood of Nolan et al. (1984) converted to plasma. These three rate constants cannot be used independently of each other since they were not unconditionally optimized. The fraction absorbed (Fa) was listed in various studies (see Annex 2, Tab. S4[2]); therefore, an average value of 0.35 was used as default.

The blood/plasma ratio of TCPy was assumed to be 1, since no methods are available to predict the blood/plasma ratio for acidic compounds. The slowly perfused tissue/blood partition coefficient was assumed to be equal to the muscle/blood partition coefficient, and the richly perfused tissue/blood partition coefficient was assumed to be equal to the kidney tissue/blood partition coefficient. The uterus (muscle tissue and endometrium tissue) is considered a slowly perfused organ (ICRP, 2002) and hence was assumed to have the same partition coefficient as the slowly perfused tissue.

[3] https://www.R-project.org/

[4] https://www.rivm.nl/en/proast

Metabolism of CPF to CPO and TCPy occurs in the liver by various CYP enzymes, while the clearance of CPO to TCPy occurs in liver and blood by serum paraoxonase-1 (PON1). The relevant *in vitro* apparent metabolic parameters (maximum metabolic rate (V_{max}) and Michaelis-Menten constant (K_m)) for each conversion, and their relevant scaling factors (e.g., intersystem extrapolation factors) were obtained from Zhao et al. (2022). The obtained parameters were further extrapolated to the corresponding *in vivo* situation based on the equation and scaling factors described in Zhao et al. (2022). See Annex 2, Table S2[2] for the V_{max} and K_m values and the equations used. The urinary excretion rate constants for CPF and CPO were set to 0 h^{-1} since no excretion of these compounds was seen in human data.

Interindividual variation in physiology and kinetics are not included in the PBK model. Interindividual variation was accounted for, combined with toxicodynamic differences, as specified in Section 2.1.9.

Model evaluation

To evaluate the performance of the developed PBK model, available human data was compared to the matched simulations. The following data was available for the evaluation: time-dependent blood or plasma concentrations of CPF (individuals), time-dependent blood or plasma concentrations of TCPy (available for individuals), and cumulative urinary excretion of TCPy (average of individuals) (Drevenkar et al., 1993; Eyer et al., 2009; Nolan et al., 1984; Smith et al., 2014; Timchalk et al., 2002). The data used for the calibration of the three absorption rate constants were not used for model evaluation (see Annex 2, Tab. S4[2] for all available data). When necessary, blood concentrations were converted to plasma concentrations using the relevant blood/plasma ratios. It was not possible to evaluate the simulations of uterus concentrations due to lack of human data, nor was it possible to evaluate the model using CPO data due to their limited quality (Timchalk et al., 2002; Eyer et al., 2009).

In the case of controlled exposure studies, simulations were run using available information reported in the studies (dose, Fa and bodyweight). Fa was used as input since the type of oral administration (in a capsule vs applying CPF in solvent on the surface of a lactose tablet) greatly influences the uptake fraction (Timchalk et al., 2002; Smith et al., 2014). When concentration-time profiles came from intoxications, this type of information was often lacking, and default values were used (bodyweight of 70 kg and Fa of 0.35). For intoxications, the dose was estimated by range-finding based on the simulated curve of the *in vivo* data. The overview of the used human studies and the relevant information (e.g., the dose, information on subjects, and the type of available data) can be found in Annex 2, Table S4[2]. Observed vs predicted plots and concentration-time profiles for each study used to evaluate the model were made using GraphPad Prism 9.3.1. In addition, the percentage of simulated datapoints that were within 2-fold, 3-fold, and 5-fold of the observed datapoints were calculated and can be found in Annex 2, Table S4[2].

Local sensitivity analysis of the PBK model

To link the simulated concentrations to the *in vitro* obtained values, the simulated output was described as the area under the curve (AUC) and the maximum concentration (Cmax) in the uterus for combined CPF and CPO (expressed as CPF using relative potency factors (RPFs)) over 10 days ($\sum AUC_{\text{CPF equivalents,uterus,0-10}}$ or $Cmax_{\text{CPF equivalents,uterus,0-10}}$) (for details on metric selection and RPF use, see Section 2.1.6 and 2.1.7, *Exposure duration and type*). As a result, the impact of individual input parameters (independent of other parameters) on the model output ($\sum AUC_{\text{CPF equivalents,uterus,0-10}}$ and $Cmax_{\text{CPF equivalents,uterus,0-10}}$) was estimated by performing a local sensitivity analysis. Normalized sensitivity coefficients (SC) were determined using Equation 1:

$$SC = \frac{(C'-C)}{(P'-P)} * \frac{(P)}{(C)} \qquad \text{(Eq. 1)}$$

in which P represents the parameter value in the PBK model, P′ represents the parameter value with a 5% increase, C is the output of the model (AUC/Cmax of CPF and CPO combined) with the original model parameter value, and C′ is the model output with 5% increase in the model parameter value (Evans and Andersen, 2000). The sensitivity analysis was carried out at a bolus intake dose of 1.8×10^{-4} mg/kg bw. This intake dose represents the 99.9th percentile of the exposure distribution for women at childbearing age (18-45 years) in the Netherlands (see Section 3.4.1).

2.1.5 Step 6: Select PBK model compartment for allocation of *in vitro* data

This case study aims to determine the external dose of CPF leading to the onset of DNT. Since the onset of DNT was determined using human NPCs mimicking early pregnancy, the *in vitro* intracellular concentrations were allocated to the uterus compartment. Even though the uterus compartment describes the whole organ, it is assumed that all of the compound will be found in the cells (and not in the interstitial fluid). At this time, it is not possible to discriminate between the cellular concentration and the interstitial fluid concentration in the PBK model since distribution parameters for CPF and metabolites are lacking.

2.1.6 Step 7: Select dose metric for extrapolation

Since reduced cell viability was assumed to be the result of prolonged exposure (10 days *in vitro* exposure), the area under the curve (AUC) was chosen as appropriate dose metric for the *in vitro* data (Groothuis et al., 2015). The $AUC_{\text{CPF, }\textit{in vitro}}$ (of the *in vitro* experiment) is calculated by multiplying the intracellular concentrations with the total exposure time (10 days × 24 h). For comparison, the Cmax was also used as a metric. This metric is considered relevant in case effects are the result of short-term exposure to a (peak) concentration ($Cmax_{\text{CPF, }\textit{in vitro}}$) and may be of interest if the DNT effects are triggered during a (short) critical time window of embryonic development.

2.1.7 Step 8: Select exposure duration and type and perform reverse dosimetry (Extrapolation C)

Exposure duration and type

To simulate the onset of DNT following CPF exposure, we considered 10 consecutive days (i.e., the time the *in vitro* experiment lasts) of oral dietary repeated exposure to be appropriate. While in the *in vitro* assay cells were exposed continuously, the choice to simulate three exposures a day at 08:00, 12:00, and 18:00 ($t = 0$, $t = 4$ and $t = 10$), instead of a single bolus exposure, stems from the different food products in which CPF is found, which are likely consumed during breakfast, lunch, and dinner.

In future research, additional refinement of the extrapolation could for some chemicals be achieved by incorporating continuous background exposure in the PBK model. However, due to the short half-life of CPF (Wołejko et al., 2022), this refinement was not foreseen to significantly affect the extrapolation, and therefore not included.

Calculation of the relative potency factor

To estimate the combined toxicity of CPF and CPO, expressed as the sum of CPF equivalents, the relative potency of CPO compared to CPF was calculated using the RPF approach (Bil et al., 2021; van den Brand et al., 2022). In the RPF analysis, a four-parameter exponential model was fitted to the continuous effect data (y; in this case the cell viability response) and plotted against the linked intracellular concentrations (x):

$$y = a\,(c^{1-e^{\left(-\left(\frac{x}{b}\right)^{d}\right)}}) \quad \text{(Eq. 2)}$$

In a covariate analysis, parallel curves were fitted to the data by applying the same shape parameters (maximum response parameter c and the steepness parameter d) in the four-parameter exponential model to both CPF and CPO, but allowing the background (parameter a), the potency (parameter b), and the residual variance to be different between compounds and experiments. For details on model fitting see RIVM (2021) and EFSA (2017). A good description of the data of each compound by the parallel curves was confirmed by visual inspection.

The RPF is defined as the ratio of two intracellular concentrations, x_{CPF} for the reference compound CPF and x_{CPO} for CPO, which both result in the same relative change in viability response (y):

$$RPF_{CPO} = \frac{x_{CPF}}{x_{CPO}} \quad \text{(Eq. 3)}$$

If the curves of the two substances are parallel, $\frac{x}{b}$ in equation 2 has the same value for both substances:

$$\frac{x_{CPF}}{b_{CPF}} = \frac{x_{CPO}}{b_{CPO}} \quad \text{(Eq. 4)}$$

Combining Equations 3 and 4 gives Equation 5:

$$RPF_{CPO} = \frac{b_{CPF}}{b_{CPO}} \quad \text{(Eq. 5),}$$

which shows that the RPF for CPO can be directly obtained from its potency parameter b and that of CPF. Calculation of the RPF-CPO and its 90% confidence interval based on potency parameter b was performed using PROAST software (version 70.3) run in R (version 4.2.0) (see [4] and RIVM, 2021). Since CPF is used as the reference compound, its RPF equals 1.

Reverse dosimetry

To determine the three times daily dietary exposure that leads to specific uterus concentrations, reverse dosimetry was performed. For the reverse dosimetry based on the AUC, the AUCs of CPF and CPO in the uterus over 10 days (the time the *in vitro* experiment lasts), taking into account the relative potency of CPF and CPO, were combined as follows:

$$\sum AUC_{CPF\ equivalents,uterus,0-10} = AUC_{\mathrm{CPF,uterus,0-10}} * RPF_{CPF} + AUC_{\mathrm{CPO,uterus,0-10}} * RPF_{CPO} \quad \text{(Eq. 6)}$$

Here, $AUC_{CPF,uterus,0\text{-}10}$ and $AUC_{CPO,uterus,0\text{-}10}$ are the AUCs of CPF and CPO in the uterus, respectively, as a proxy of the 10-day exposure in the experiment.

For the reverse dosimetry based on the Cmax, Equation 7 was used to combine the Cmax of CPF and CPO (expressed as CPF equivalents) in the uterus over the 10 days the experiment lasted:

$$Cmax_{CPF\ equivalents,uterus,0-10} = \max\left(C_{\mathrm{CPF,uterus(t)}} * RPF_{CPF} + C_{\mathrm{CPO,uterus(t)}} * RPF_{CPO}\right) \quad \text{(Eq. 7)}$$

Here, $C_{CPF,uterus(t)}$ and $C_{CPO,uterus(t)}$ are the concentrations of CPF and CPO in the uterus, respectively, over time during 10 days of exposure.

The daily oral doses potentially leading to the onset of DNT, in mg/kg bw/day, were determined by setting $\sum AUC_{CPF\ equivalents,uterus,0\text{-}10}$ (and $Cmax_{CPF\ equivalents,uterus,0\text{-}10}$) equal to the *in vitro* $AUC_{CPF,\ in\ vitro}$ (and $Cmax_{CPF,\ in\ vitro}$), obtaining novel external dose-response relationships.

2.1.8 Step 9: Select benchmark response for benchmark dose modelling

Using the extrapolated external doses based on cell viability effect concentrations (using the dose metrics $AUC_{CPF,\ in\ vitro}$ and $Cmax_{CPF,\ in\ vitro}$) allocated to the uterus, BMD analysis was carried out. A benchmark response (BMR) of 5% decrease in cell viability compared to the control group was applied in the BMD analysis, because 5% is the default value describing the change in mean response compared to the controls (EFSA, 2017). Again, PROAST software (version 70.3) run in R (version 4.2.0) was used for the analysis.

In the BMD analysis, four dose-response models were fitted, namely the exponential model, the Hill model, the inverse exponential model, and the lognormal model. The fitting of each model results in a BMD corresponding with the BMR of 5% (BMD_{05}) as well as the 90% confidence interval of the BMD. The lower and upper 95% confidence limits of the BMD_{05} are $BMDL_{05}$ and $BMDU_{05}$, respectively. To establish the final BMD_{05} confidence interval based on the four fitted models, results from these models were combined by taking the lowest $BMDL_{05}$ and the highest $BMDU_{05}$ from these four models. In the analysis, possible differences between experiments with respect to the background response and the residual variation were taken into account. This was done by applying the covariate approach in the BMD analysis, as previously described (Section 2.1.7, *Calculation of the relative potency factor*).

2.1.9 Apply assessment factors (Extrapolations D, E and F) to obtain an *in vitro*-based HBTV

Since neuronal progenitor cells are derived from human embryonic cells, no interspecies extrapolation is needed (Extrapolation D). The extrapolation from short-time exposure *in vitro* to chronic daily exposure is elaborated in Section 4 (Extrapolation E).The extrapolation from the average human to a sensitive human (Extrapolation F) was performed by manually applying an assessment factor (AF_{intra}) of 10 (EFSA, 2012).

2.2 Exposure and risk assessment

To illustrate the use of an *in vitro*-based HBTV, a theoretical example of a NAM-based risk assessment was performed. The risk assessment was based on the calculated *in vitro*-based HBTV and calculated dietary exposure.

2.2.1 Dietary exposure assessment

Dutch food consumption data were collected in the Dutch National Food Consumption Survey from 2007 to 2010[5]. This survey consists of two non-consecutive 24-hour dietary recalls. The food consumption data were coded with the food coding system FoodEx1, and included into EFSA's Comprehensive Database (EFSA, 2011). Since the occurrence data were reported for the raw primary commodities (RPC), the consumption data for composite foods or RPC derivatives (i.e., a single-component RPC altered by processing) need to be converted to their equivalent quantities of RPCs. To this end, the RPC model was applied by EFSA, resulting in the RPC Consumption Database (EFSA, 2019c), from which the relevant consumption data were extracted by EFSA upon approval of the data owner. Women of childbearing age (18-45 years) were selected from the survey (N = 485) as a proxy for pregnant women.

Occurrence data from pesticide residues were obtained from the official monitoring activities carried out in the European Union, United Kingdom, Iceland, and Norway[6]. Data over the years 2014-2018 were downloaded from the Zenodo database[6] for 29 European countries. The occurrence data set was restricted to 30 RPCs of interest for the exposure assessments as defined for the studies by EFSA (2019b,d) and van Klaveren et al. (2019a,b). Only measurements obtained from selective or objective sampling and referring to CPF were used, and the data from all countries were pooled into one single data set.

Exposure calculations were performed probabilistically using the Monte Carlo Risk Assessment (MCRA) tool version 9.1 (MCRA[7]). Chronic (long-term) exposure was calculated using the observed individual means (OIM) model. Further criteria and assumptions were according to "Tier II", as established at the European level in 2018 and applied and described by EFSA (2019b,d), van Klaveren et al. (2019a,b), and Te Biesebeek et al. (2021). Input data regarding authorized uses and maximum residue levels (MRLs) applicable in the European Union were updated to the situation on 31-12-2018. Hence, MRLs were extracted from the EU Pesticides Database[8], and information on authorized uses was deduced from EFSA's MRL-review of CPF (EFSA, 2017). A bootstrapping approach was used to quantify sampling uncertainty in food consumption and concentration data caused by a limited sampling size (Efron and Tibshirani, 1993). This approach resamples (with replacement) the original food consumption and concentration dataset to obtain a bootstrap of n observations. In the present calculation, an uncertainty analysis was performed using 100 resample cycles with 10,000 iterations each. This yielded 100 alternative exposure distributions, which might have been obtained during sampling from the population of interest and during sampling of foods. Of those 100 alternative exposure distributions, the p50 of the percentile of interest together with the 95% uncertainty interval was obtained. Within the European pesticide framework, the threshold for regulatory consideration agreed among Member States is at the 99.9th percentile of the exposure distribution. Therefore, the 99.9th exposure percentile together with the 95% confidence interval was taken as the percentile of interest for the risk assessment.

2.2.2 Risk assessment

Risk assessment was performed by comparing the calculated dietary oral exposure to the *in vitro*-based HBTV. A ratio equal to or lower than 1 indicates that no appreciable adverse health effects are expected. The calculated results in this paper should be seen as part of this proof-of-principle study and not as the outcome of a full risk assessment (see Section 4).

3 Results

3.1 Obtaining the *in vitro* concentration-response relationship, expressed as intracellular concentration

We observed a correlation between transcriptomics of the hNPT at day 10 (normalized to undifferentiated NPCs) and the human brain at different (gestational) ages (normalized to certain developmental periods) (Fig. S1[2]). The results show that the up- and

[5] https://www.rivm.nl/bibliotheek/rapporten/350050006.pdf
[6] https://zenodo.org/
[7] https://mcra.rivm.nl/
[8] https://food.ec.europa.eu/plants/pesticides/eu-pesticides-database_en

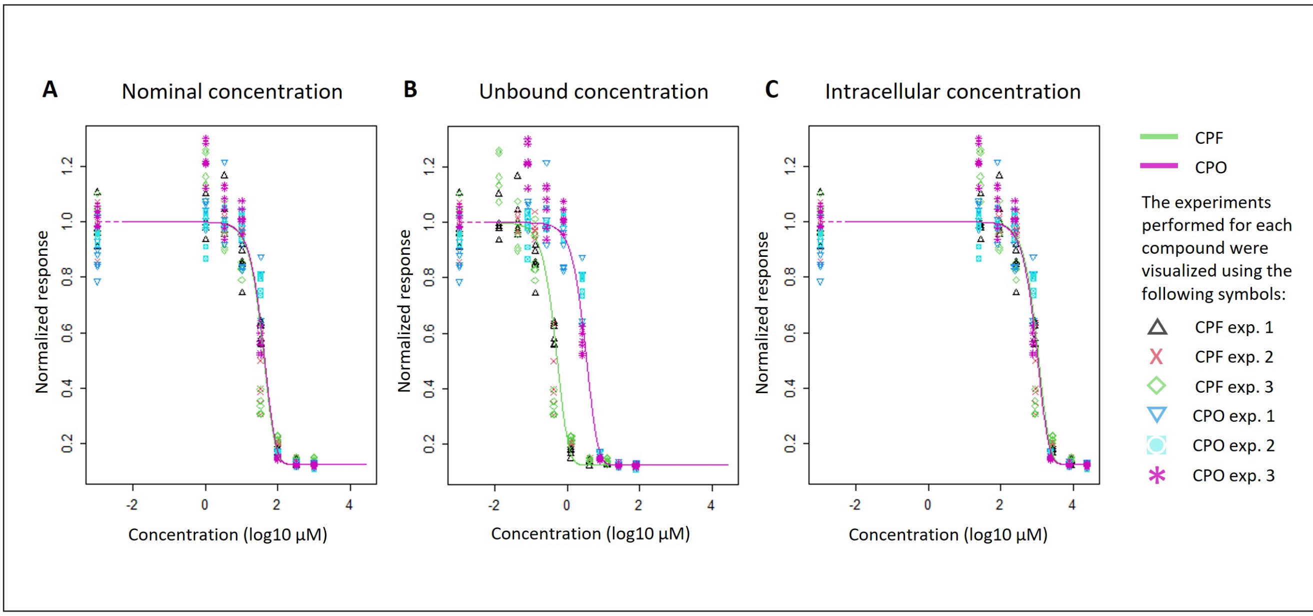

Fig. 3: Cell viability in the hNPT following 10 days exposure to CPF (green) or CPO (purple)
Viability is normalized to the modeled background response for each substance and experiment and plotted against the log nominal concentration (A), the calculated log free concentration in medium (B), and the calculated log intracellular concentration (C) for 6 replicates in each experiment (n = 6) and N = 2-3 experiments. (Individual) values can be found in Annex 1, Table S1[2].

downregulated genes in the hNPT at day 10 correlate with the brain at approximately 8-10 to 16-18 weeks of gestation, indicating that the hNPT mimics neuronal-astroglial differentiation during early pregnancy.

To determine the effects of CPF and its metabolite CPO on cell viability, differentiating human NPCs were exposed for 10 days. As can be seen in Figure 3, CPF and CPO reduce cell viability in the hNPT. Looking at nominal concentrations (Fig. 3A), CPF and CPO are seemingly equipotent in the reduction of cell viability.

Based on the nominal concentrations and the assumption that lipophilicity mostly drives *in vitro* distribution, the free concentrations in medium and the intracellular concentrations of CPF and CPO were estimated for the whole exposure duration. Assuming no saturation of binding to serum constituents, plastic and cells, it is estimated that CPF is mostly bound to medium constituents. Only 1% is predicted to be freely available in medium. The Kramer (2010) model predicts negligible binding to well plate plastic (5%) and accumulation in cells (1%). The predicted distribution of CPO differs only slightly from CPF in that 4% binds to plastic and 8% is unbound in medium. Similar to CPF, 1% of CPO is calculated to be found in cells, and most of CPO is bound to medium constituents. Given the low accumulation levels in cells, no significant accumulation of CPF and CPO after refreshment of exposure medium was seen in additional calculations (data not shown). Note that the low uptake of CPF and CPO into cells results in relatively high intracellular concentrations (Fig. 3C) due to the small volume of the cells.

Figures 3B and 3C show the same viability response, plotted against the calculated free concentrations of CPF and CPO in medium and the intracellular concentrations (cell-associated amount divided by cellular volume), respectively. Depending on the choice of concentration metric, which can depend on the research question, CPF seems most potent when considering free concentrations in medium and least potent when considering intracellular concentrations. In our case, we considered intracellular concentrations to be the appropriate concentration metric as we assumed that the inhibition of the metabolic activity of differentiating NPCs is directly related to intracellular concentrations of CPF and CPO. Expressing concentrations as nominal, unbound or intracellular concentrations (Fig. 3A-C), CPF and CPO are (seemingly) equipotent in the reduction of cell viability when expressed against nominal and intracellular concentrations, yet not unbound concentrations.

3.2 Obtaining the *in vitro*-based HED

3.2.1 PBK model

Evaluation against in vivo *concentration-time curves*
The performance of the PBK model built in the present study was evaluated against available *in vivo* human data not used for the calibration of absorption parameters. Figure 4 shows that the developed PBK model adequately predicts the available human data in both plasma and urine for CPF and its metabolite TCPy. Since the Fa turned out to depend greatly on the CPF administration type (range of 0.1-0.8 due to administration via a capsule vs applying CPF in solvent on the surface of a lactose tablet) (Timchalk et al., 2002; Smith et al., 2014), we used the reported Fa values from

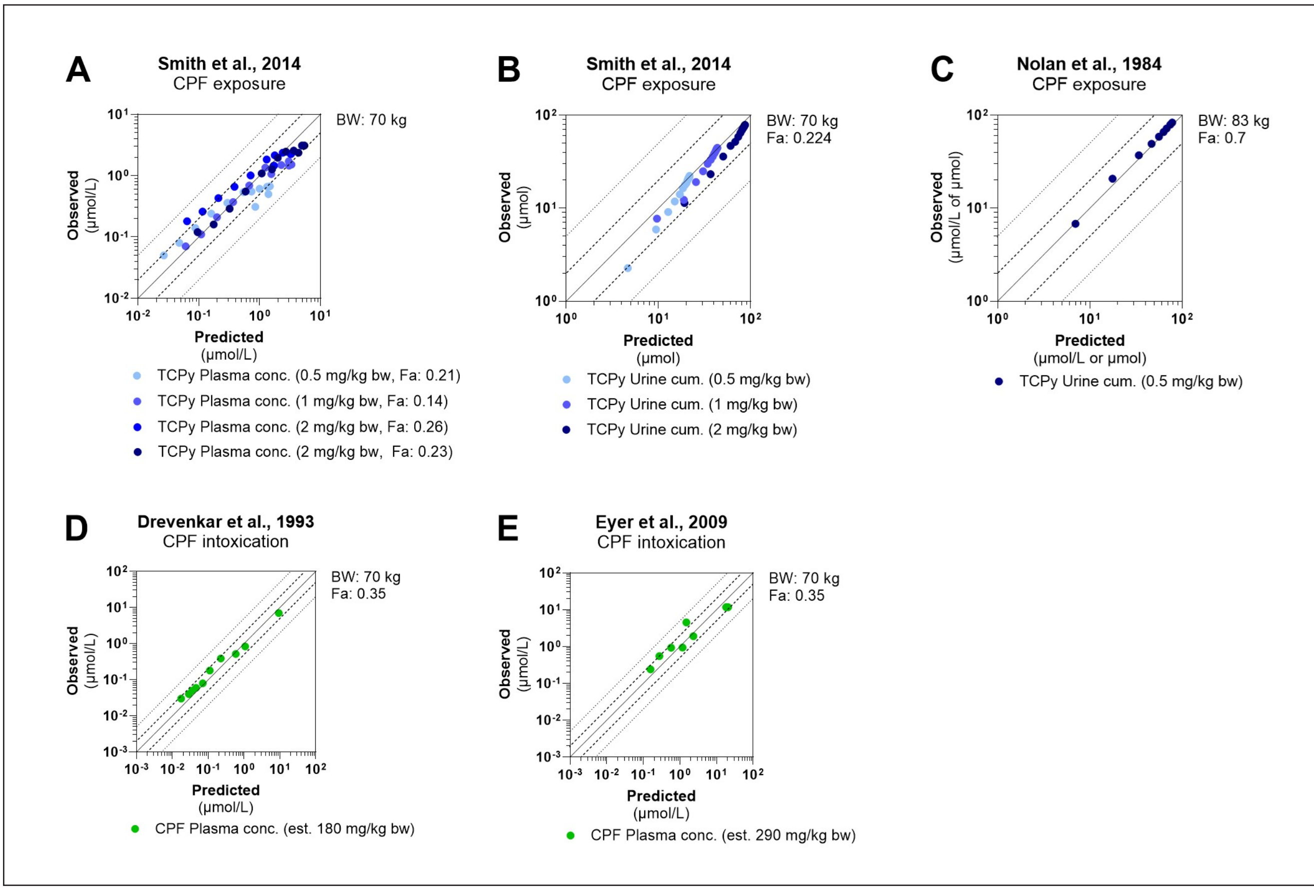

Fig. 4: Comparison of the observed human data from *in vivo* studies and predicted values from the model
The data points were plotted in a graph with double logarithmic scale. The lines visualize no difference (solid lines), 2-fold variation (striped lines), and 5-fold variation (dotted lines) between the simulated and measured data points. TCPy was measured following experimental exposure (A-C), while CPF was measured in intoxication studies (D and E). The used bodyweights, fractions of absorption, and (estimated) doses are listed next to the graphs. More information on the studies can be found in Annex 2, Table S4[2]. For graph C, the data was obtained from Smith et al. (2014), which is originally from Nolan et al. (1984).

the individual studies in the evaluation when they were available. Otherwise, we used the average Fa value of 0.35 (see Annex 2, Tab. S4[2]).

Of all available data points obtained from literature, 89% and 99% fall within 2-fold or 3-fold difference from the simulated data points, respectively. When differentiating between the simulated plasma concentrations and cumulative urine amounts, 98% of cumulative urine amount simulations are within 2-fold whilst 83% of the plasma concentrations fall within 2-fold of the *in vivo* data. When evaluating the simulations for CPF and CPO separately, 90% of the CPO simulations and 89% of CPF simulations are within 2-fold of the *in vivo* data. The overview of the concentration/amount-time graphs, including both simulations and *in vivo* data, can be found in Annex 2, Figure S2A-K[2].

Two available studies were excluded from evaluation of the model due to the limited quality of the data. Timchalk et al. (2002) published concentration-time relationships of CPF in plasma with very few data points very early in time. Additionally, no dose-dependent increase in CPF plasma levels was seen. Eyer et al. (2009) published the CPO concentration in plasma following intoxication. The concentration-time curve showed an irregular shape in which two peaks in concentration were seen that could not be explained.

Sensitivity analysis of PBK model parameters

Figure 5 shows the results of the sensitivity analysis reflecting the impact of each PBK input parameter on the model output of interest ($\sum AUC_{CPF\ equivalents,uterus,0\text{-}10}$ or $Cmax_{CPF\ equivalents,uterus,0\text{-}10}$) following a daily oral exposure to CPF at 1.8×10^{-4} mg/kg bodyweight for 10 days (see Section 3.4.1). Only the parameters with a normalized sensitivity coefficient higher than 0.1 (absolute value) are shown in Figure 5. Both $\sum AUC_{CPF\ equivalents,\ uterus,0\text{-}10}$ and $Cmax_{CPF\ equivalents,\ uterus,0\text{-}10}$ are significantly affected by the fraction liver tissue of body weight, fraction absorbed, scaling factor of human liver microsome in mg to g liver, and the dose. In contrast, less sensitive parameters are sev-

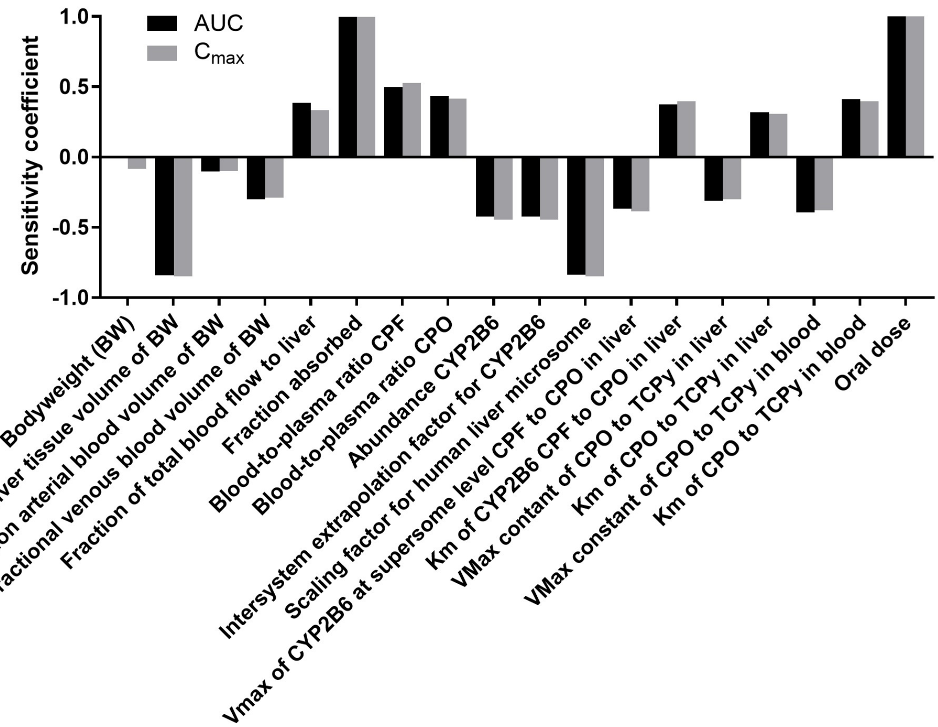

Fig. 5: Sensitivity analysis of PBK model parameters for the prediction of the summed equivalent intracellular concentration of CPF and CPO in the uterus upon a daily oral CPF dose of 1.8 x 10^{-4} mg/kg bodyweight (representing the 99.9th percentile of the exposure distribution of women at childbearing age (18-45 years) in the Netherlands, see Results section 3.4.1) for 10 days using either AUC or Cmax as dose metric

eral physiological parameters (i.e., fraction arterial blood of body weight, fraction venous blood of body weight, fraction of blood flow to liver), physicochemical parameters (e.g., blood/plasma ratio of parent, blood/plasma ratio of CPO), and metabolic parameters (e.g., abundance of CYP2B6, intersystem extrapolation factor of CYP2B6, V_{max} and K_m of CYP2B6 for pathway 1, and V_{max} and K_m for pathways 3 and 4). Body weight was found to have limited impact on $Cmax_{CPF\ equivalents,0\text{-}10}$ but showed no effect on $\sum AUC_{CPF\ equivalents,\ 0\text{-}10}$.

3.2.2 RPF analysis

The RPF for CPO was derived using CPF as the reference compound. The analysis was performed using the intracellular concentration-response data for cell viability of the hNPT. Intracellular concentrations were used since these are assumed to be most relatable to uterus tissue concentrations. Analysis on the intracellular concentration-response relationship showed a difference in background response between plates (see parameters “a” in Fig. 6) and residual variance between compounds and experiment (see parameters “var” in Fig. 6), which were taken into account in the RPF analysis. The potency did not differ between both substances and experiments, i.e., parameter b (which characterizes the potency of a compound) is the same for different substances and experiments. As a consequence, the derived RPF_{CPO} of 1.077 [90% CI of 0.99-1.15] (Fig. 6) is very close to 1, and the CI overlaps with 1, and it cannot be concluded that CPF and CPO differ in potency. The data used to perform the RPF analysis are provided in Annex 4, Table S6[2].

3.2.3 Reverse dosimetry and BMD analysis

The intracellular concentrations were extrapolated using both the AUC and Cmax dose metrics (Section 2.1.7; Tab. 1). These extrapolations resulted in two external dose-response curves (Tab. 1).

The BMD analysis was carried out on these external dose-response curves to derive the 90% confidence interval (CI) of the BMD_{05}. For the AUC and Cmax dose metrics, the BMD_{05} confidence intervals are 224-450 mg/kg bw/day and 143-314 mg/kg bw/day, respectively. Since the BMD_{05} CIs overlap, it is not possible to say that the choice of dose metric induces different BMD_{05} values. The *in vitro*-based HED is set equal to the lower confidence interval of the BMD_{05}, the $BMDL_{05}$. Full details of the analysis, *in vitro* AUC and Cmax values, and external values based on $AUC_{CPF\ equivalent,uterus,0\text{-}10}$ and $Cmax_{CPF\ equivalent,uterus,0\text{-}10}$ are

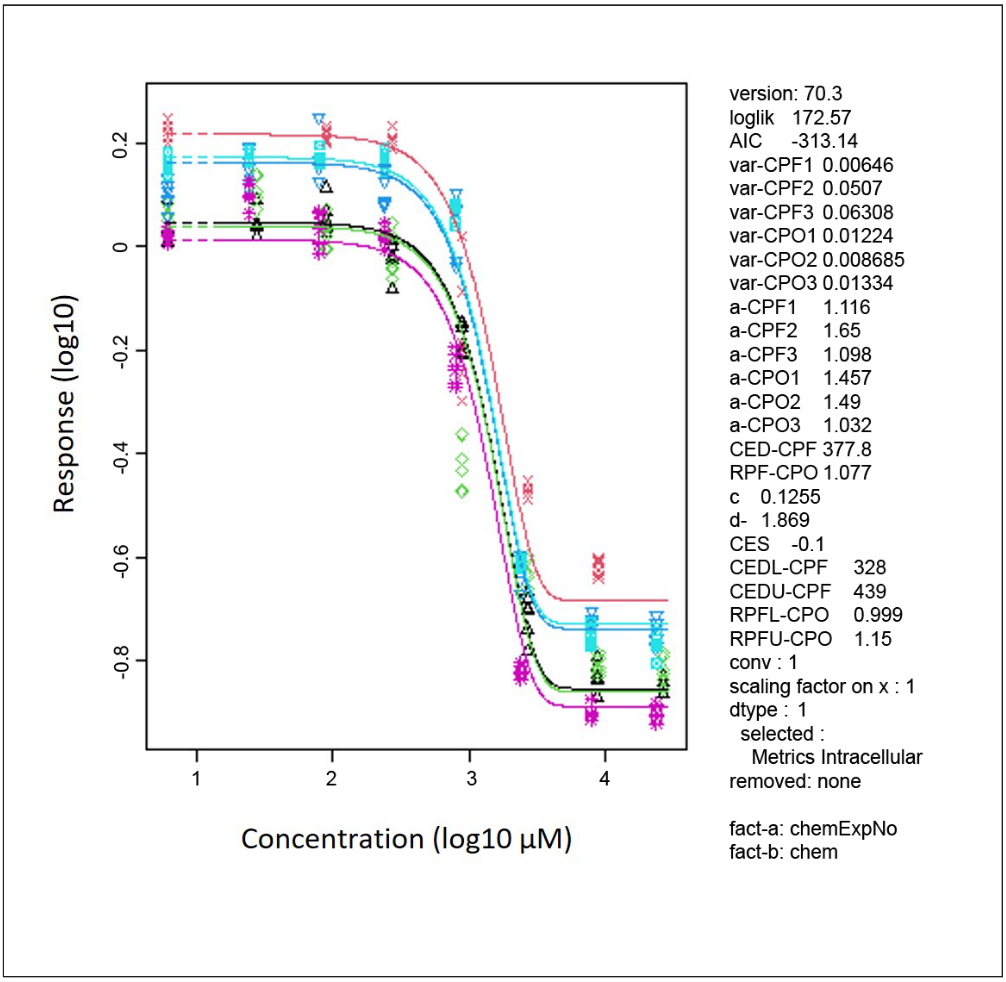

Fig. 6: RPF analysis for CPO, with CPF as index compound, based on the intracellular concentration-response curve showing effects on cell viability following 10 days of exposure

provided in Annex 5[2]. Simulations of the combined AUC and Cmax for CPF and CPO in the uterus at $BMDL_{05}$ and BMDU05 are shown in Figure 7.

3.3 Obtaining the *in vitro*-based HBTV

To convert an *in vitro*-based HED into an *in vitro*-based HBTV, at least two additional extrapolation steps need to be performed: 1) to extrapolate between the different exposure durations *in vitro* and *in vivo* (Extrapolation E), and 2) to extrapolate between sensitive populations (intraspecies extrapolation, Extrapolation F). Unfortunately, it is not yet possible to quantify the uncertainty of Extrapolation E (see Section 4). For Extrapolation F, an AF_{intra} of 10 was included to calculate *in vitro*-based HBTVs of 22 mg/kg bw/day (based on AUC) and 14 mg/kg bw/day (based on Cmax).

3.4 Risk assessment based on *in vitro*-based HBTV

3.4.1 Dietary assessment

The P99.9 exposure to CPF for women at childbearing age (18-45 years) in the Netherlands is 1.8×10^{-4} mg/kg bw/day (95% confidence interval: $1.6\text{-}2.0 \times 10^{-4}$ mg/kg bw/day).

3.4.2 Risk assessment

The dietary exposure is several orders of magnitude lower than the *in vitro*-based HBTVs. Based on this provisional NAM-based risk assessment, dietary exposure is thus not expected to result in a decrease in the viability of neuronal progenitor cells, as a proxy for DNT, in children of Dutch women.

4 Discussion

The aim of the current paper is to present a proof-of-principle NAM-based risk assessment for the onset of DNT, independent of cholinesterase inhibition, using CPF and its metabolite CPO as an example. We aimed to additionally highlight the hazard characterization steps and extrapolations of importance when using *in vitro* and *in silico* tools for risk assessment. In the calculation of the *in vitro*-based HGBVs/HBTVs, extrapolations related to the concentration metric (Extrapolation B; nominal vs free in medium vs intracellular concentrations), reverse dosimetry (Extrapolation C; including exposure duration and type), as well as inter- and intraspecies differences (Extrapolations D and F) were quantified and included. However, we were not able to quantify extrapola-

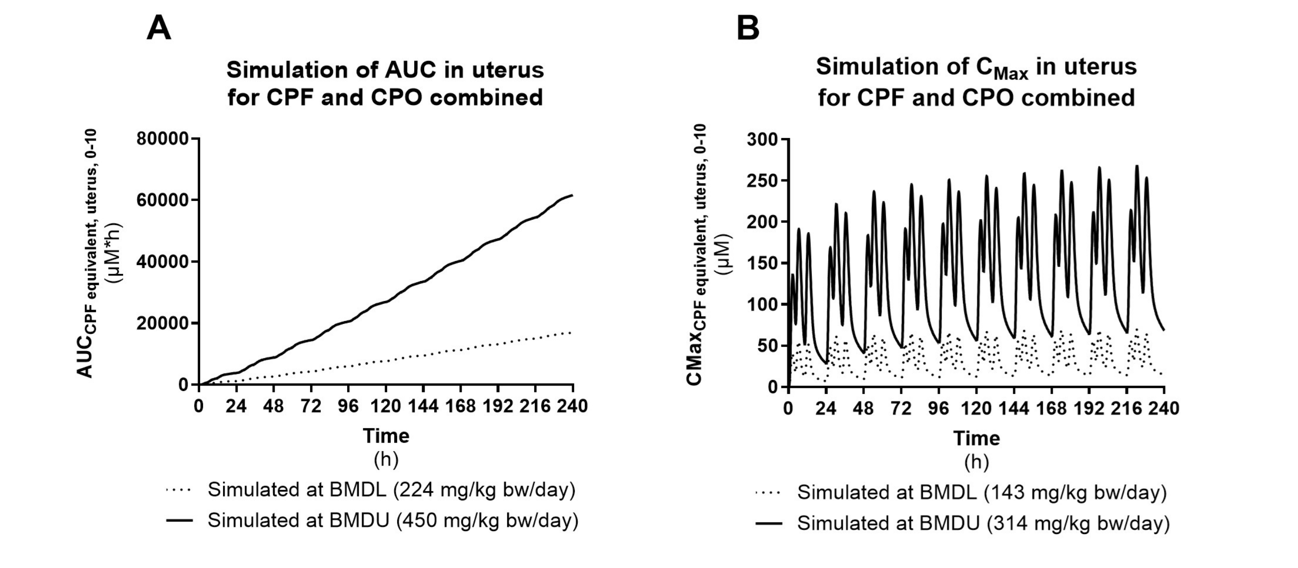

Fig. 7: Simulations of the $AUC_{CPF\ equivalent,uterus,0\text{-}10}$ and $Cmax_{CPF\ equivalent,uterus,0\text{-}10}$ at the calculated $BMDL_{05}$-$BMDU_{05}$ doses
The dotted line shows the simulation at $BMDL_{05}$, while the solid line shows the simulation at $BMDU_{05}$. Simulations were performed for a woman of 70 kg with a Fa of 0.35.

Tab. 1: Input values for and output values of the reverse dosimetry of CPF based on the cell viability of hNPT following 10 days exposure to CPF expressed as intracellular concentrations
The $AUC_{CPF,\textit{in vitro}}$ was calculated by multiplying the intracellular concentrations times 24 h and 10 days, whereas the $Cmax_{CPF,\textit{in vitro}}$ was just set to the intracellular concentrations. The $AUC_{CPF,\textit{in vitro}}$ was set to the $AUC_{CPF\ equivalent,\ uterus,0\text{-}10}$ and the $Cmax_{CPF,in\ vitro}$ was set to the $Cmax_{CPF\ equivalent,\ uterus,0\text{-}10}$ to obtain external doses related to the intracellular concentrations *in vitro*.

Chlorpyrifos						
Cell viability of hNPT			**Dose metric, expressed as**		**External dose, based on**	
Average	SD	Intracellular concentration	$AUC_{CPF,\textit{in vitro}}$	$Cmax_{CPF,\textit{in vitro}}$	$AUC_{CPF\ equivalent,\ uterus,0\text{-}10}$	$Cmax_{CPF\ equivalent,\ uterus,0\text{-}10}$
%	%	µM	µM×h	µM	mg/kg bw/day[a]	mg/kg bw/day[a]
109	9.9	27	6517	27	131.5	82
100	5.7	90	21508	90	254	163.5
89	8.4	272	65175	272	461	313.5
39	15	896	215076	896	854	614.8
8.4	3.2	2716	651746	2716	1556.5	1124.5
1.8	1.9	8962	2150763	8962	3509	2336.3
0.38	1.7	27156	6517463	27156	8983	5602

[a] Total of 3 equal doses mimicking eating occasions at 08:00, 12:00, and 18:00.

tions between the key event and the adverse outcome (Extrapolation A) and extrapolations related to differences in exposure duration *in vitro* and *in vivo* (Extrapolation E).

Regarding Extrapolation A, it is not yet possible to quantitatively link the activation of the key event "Increase, Cell injury/death" (represented by the viability of differentiating NPCs) to the adverse outcome indicative of DNT. This is of importance since thresholds between key events and feedback mechanisms could decrease progress to the adverse outcome. Neurodevelopmental effects in children linked to maternal CPF exposure (as listed in Section 1) are all linked to the adverse outcomes presented in Spinu et al. (2019) (i.e., "Impairment of Learning and Memory and Cognitive Function, Decreased", "Neurodegeneration" and "Parkinsonian motor deficits"), and the key event "Increase, Cell injury/death" plays a central, albeit indirect, role in these AOPs. To quantitatively link the key event "Increase, Cell injury/death" to the onset of DNT-related adverse outcomes, a DNT-specific quantitative adverse outcome pathway (qAOP), through which CPF may act, is needed. Spinu et al. (2022) proposed a simplified qAOP for DNT based on three linked key events, but did not include "Increase, Cell injury/death". The included key events were studied independently, and all had a binary (yes/no) readout, thus not allowing the determination of a continuous quantitative relation between the key events and the onset of DNT. While reduction of cell viability may be considered a non-specific endpoint for the onset of DNT, we have measured it in a cell type relevant for DNT. The reduction in cell viability by CPF (and other compounds) is related to the disturbance of relevant neurodevelopmental processes in the hNPT measured on the level of gene expression changes (de Leeuw et al., 2022). Therefore, cell viability was considered a suitable endpoint in this proof-of-principle. Other researchers have measured the effect of CPF on neural-specific endpoints, such as synaptogenesis, neuronal outgrowth, and BDNF levels, but only found significant effects at concentrations also affecting cell viability (Di Consiglio et al., 2020; Algharably et al., 2023). In previous research, we found that neurite area was equally sensitive to CPF exposure as cell viability (de Leeuw et al., 2022). Furthermore, it should be noted that our IC5 for CPF is similar and sometimes lower than concentrations at which effects were seen by Algharably et al. (2023) and assays of the DNT *in vitro* battery (Masjosthusmann et al., 2020). For future research, neural network formation may be an interesting endpoint to further explore in the hNPT as this is the most sensitive endpoint in the DNT *in vitro* battery (Masjosthusmann et al., 2020). A final interesting result of Extrapolation A is that, when expressed against nominal and intracellular concentrations, CPF and CPO are seemingly equipotent. We can conclude from these results that while CPO may more potently inhibit AChE compared to CPF (Monnet-Tschudi et al., 2000), it does not inhibit cell viability of hNPT more potently.

For Extrapolation B, we assumed that the inhibition of the metabolic activity of differentiating NPCs, as proxy of reduced cell viability, is induced by intracellular concentrations of CPF and CPO. Therefore, we calculated the distribution of CPF and CPO in the *in vitro* system based on the lipophilicity of the compounds, the type of cells, the type of medium, and the type of well plate used. We estimated that intracellular concentrations are approximately 27-fold higher than the applied nominal concentrations, whereas estimated free concentrations in medium are approximately 80-fold lower than the applied nominal concentrations. This indicates that the choice of the concentration metric could drastically affect the calculation of the *in vitro*-based HGBV. It must be noted that the model for predicting the free and intracellular concentrations was developed based on a set of polycyclic aromatic hydrocarbons (PAHs) (Kramer et al., 2010). While additional research needs to verify model predictions for other chemical classes, CPF and CPO are, like PAHs, nonpolar compounds with similar Log Kow values (Log Kow of PAHs used for model calibration was 3.33-6.13; Log Kow is 4.784 for CPF and 3.894 for CPO; see Tab. S2[2]) (Kramer, 2010). However, the predicted intracellular concentration could be over- or underestimated, and additional research on the applicability domain and uncertainty of the different models used in the extrapolation should be performed.

For the extrapolation of internal concentrations to external doses (Extrapolation C), we first developed a PBK model to allow simulation of CPF, CPO, and TCPy kinetics in women of childbearing age (18-45 years). The developed PBK model predicted human plasma and/or cumulative urinary concentrations from five studies well, with 89% and 99% of all predicted values deviating less than 2- or 3-fold, respectively, from the *in vivo* values. To apply reverse dosimetry, the obtained *in vitro* data were attributed to the uterus compartment, as the *in vitro* assay mimics early pregnancy (approximately week 8-10 to week 16-18). Since no placental or other barriers are present at this stage (Tetro et al., 2018; Turco and Moffett, 2019), it is assumed that the embryo is exposed to similar concentrations as found in the uterus. Whilst the embryo is embedded in the endometrium, the current model does not discriminate between the muscle layers, the epithelial cells lining the cavity, and the endometrial cells in which the embryo is implanted, due to lack of data describing the distribution of chemicals among these cell types. It remains unclear whether the absence of these fetal sub-compartments in the model hampers the quality of the fetal exposure estimation. Since no *in vivo* determined or *in silico* estimated partition coefficients from blood or plasma to uterus tissue specifically were available, we used the *in silico* estimated partition coefficient of slowly perfused tissue, as the uterus is considered a slowly perfused organ (ICRP, 2002). Concentrations of CPF and CPO in the uterus compartment were summed using RPF analysis, based on the *in vitro* intracellular concentration-response relationships. Due to lack of data on combined exposure, it was assumed that CPF and CPO have an additive effect on cell viability.

The choice of the most suitable dose metric for the extrapolation of internal concentrations to external doses depends on the research question, the mode of action, and the tested endpoint. Cmax is most often used since most *in vitro* assays show acute toxicity following a bolus exposure. Additionally, this dose metric is suitable for reversible toxic effects since peak concentrations may not allow for damage accumulation over time (Groothuis et al., 2015). However, based on the used *in vitro* data (prolonged exposure – 10 days) and the comparison to chronic daily exposure, the AUC may be a more appropriate dose metric as it better captures time-dependent (irreversible) cumulative effects (Groothuis et al., 2015) such as DNT. The present study performed the reverse dosimetry for 10 days of oral exposure during a critical developmental window using both dose metrics, and we only found a small, insignificant difference in estimated external daily exposure doses (expressed as $BMDL_{05}$). Inhalation exposure was not considered but may be relevant for individuals working in the agricultural sector or residing in rural areas, due to pesticide drift (Rahman, 2021). Given the uncertainties around pesticide drift, it was not feasible to take this into account. Occupational exposure was not considered because only a small percentage of Dutch farmers is female[9]. Care should be taken when extrapolating the results to individuals with high ambient exposure.

For interspecies extrapolation (Extrapolation D) often a default assessment factor of 10 is used (EFSA, 2012). However, since the cells used in this study are derived from human embryonic cells, no interspecies extrapolation was needed.

Regarding the difference in exposure duration *in vitro* and *in vivo* (Extrapolation E), in general, acute (short-term; hours, days) effects are easier to capture in *in vitro* assays than chronic (long-term; months and years) effects. In case of chronic effects and the derivation of a chronic HBGV/HBTV, the effects of long-term exposure need to be estimated. Additional research is needed to determine whether *in vitro* assays can detect markers of long-term effects, and to quantify the uncertainty related to the extrapolation of short-to-long exposure duration. The correlation between transcriptomics of the hNPT after 10 days of differentiation and the human brain at different gestational ages indicates that the 10-day differentiation *in vitro* represents *in vivo* brain development of approximately 8 weeks (from week 8-10 to 16-18). For simplicity, we assumed that neurological effects are caused by short-term exposure during a critical time window during the development of the fetus in early pregnancy (Silbereis et al., 2016). Furthermore, we assumed that cell differentiation in the hNPT *in vitro* during 10 days of incubation represents this critical time window. Thus, an extrapolation for difference in duration *in vitro* and *in vivo* is not needed.

Regarding the intraspecies extrapolation from the average human to a sensitive human (Extrapolation F), the developed PBK model currently does not consider possible interindividual variation in physiology and metabolism, i.e., it provides a description of an average woman of childbearing age. As previously indicated by Zhao et al. (2022), large variations in metabolism of CPF were observed within different Caucasian individuals, (partly) due to interindividual differences in the abundance of enzymes involved in the metabolism of CPF. By taking into account quantitative data on interindividual variation it is possible to develop chemical-specific assessment factors instead of using the default assessment factor (AF_{intra}) of 10 (EFSA, 2012). Additionally, by incorporating the uncertainty of chemical-specific parameters and physiological parameters in the simulations, the model can also quantify uncertainty related to PBK parameters (for example, see Kasteel et al., 2021).

In this study, the extrapolation from the *in vitro* concentration-response curve to external doses (*in vitro*-based HED) expected to induce the onset of DNT led to doses of 140 and 220 mg/kg bw/day, based on Cmax and AUC, respectively, and subsequently *in vitro*-based HBTVs of 14 and 22 mg/kg bw/day. Data from epidemiological studies suggest that neurodevelopmental effects in infants occur at maternal cord blood CPF concentrations between 0.005 and 7.33 ng/mL (Chiu et al., 2021; Rauh et al., 2006; 2011; Silver et al., 2017). By applying reverse dosimetry, assuming that uterus concentrations represent cord blood concentrations (assuming no barriers), we calculated corresponding maternal doses of 0.00026 to 0.38 mg/kg bw/day. These estimated maternal doses are several orders of magnitude lower than the *in vitro*-based HBTVs. This indicates that the *in vitro* endpoint might not be sensitive enough.

Since CPF is known to induce adverse outcomes linked to DNT in humans, perhaps CPF and CPO are also able to disturb other (non-cholinesterase-dependent) events in the AOPs from Spinu et al. (2019), which are not represented in the hNPT. Further research should focus on discovering and optimizing more DNT-specific key events that are independent of cholinesterase inhibition, e.g., network formation and function, and the quantitative key event relationship with DNT. Di Consiglio et al. (2020) measured the effect of CPF exposure on *in vitro* neurite outgrowth, synaptogenesis, and BDNF levels, but the predictive values of these non-cholinesterase related endpoints turned out to be low in this study, warranting further research in finding suitable endpoints and test systems (Algharably et al., 2023).

Algharably et al. (2023) also predicted oral maternal CPF exposure during pregnancy, leading to fetal brain concentrations inducing non-cholinesterase-dependent effects. A PBK model including maternal and fetal compartments, simulating 15 weeks of gestation, was combined with cell lysate concentrations obtained *in vitro* using human induced pluripotent stem cell-derived neuronal stem cells undergoing differentiation towards neurons and glia following 14 days of exposure to CPF, comparable to the hNPT used in this study. Similar to our results, Algharably et al. (2023) estimated that maternal doses of 3 to 271 mg/kg bw/day are expected to induce DNT-related effects. However, it must be noted that the reverse dosimetry applied by Algharably et al. (2023) of the rather narrow nominal concentration range used in the *in vitro* experiment (~2-fold; from 18.45 µM to 37.1 µM) resulted in an unexpected, much larger range of maternal (external) doses (> 500-fold; from 8.9 mg/kg bw/day to 5,000 mg/kg bw/day). Unfortunately, their PBK model code is not available to check and reproduce these findings.

Several differences in the qIVIVE approach are seen between the study by Algharably et al. (2023) and this study. These are, amongst others, 1) PBK model compartment for allocation of *in vitro* data (uterus vs fetal brain), 2) *in vitro* concentration range

[9] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Farmers_and_the_agricultural_labour_force_-_statistics (accessed 31.07.2025)

(1-1000 µM vs 18-37 µM), 3) inclusion of metabolites (CPO vs none), 4) endpoints (viability vs neuronal cell development and function), 5) dose metric (AUC and Cmax vs average concentration), 6) BMR (5% vs one standard deviation), 7) *in vitro*-based HED (BMDL vs BMDU). Because several important steps and extrapolations could not be retrieved, it is difficult to make a conclusive comparison between our approach and the approach of Algharably et al. (2023). It would be of great interest for the advancement of this field to study these differences in approaches and their influence on the outcome, and come to a consensus on how to perform a NAM-based risk assessment.

Using our PBK model, concentrations of CPF and its metabolites can be accurately simulated in various tissues and fluids. Even though the oral doses extrapolated based on the *in vitro* data are comparable to the doses extrapolated by Algharably et al. (2023), the doses are several orders of magnitude higher than the maternal exposure linked to DNT effects and the daily exposure of the Dutch population. It appears that differentiating NPCs might not react with the same sensitivity as cells *in vivo*, or key events of the AOP for DNT are not present in hNPCs, showing the need for an *in vitro* testing battery and qAOPs.

The current study is an exploration of NAM-based risk assessment for the onset of DNT and shows the importance of the steps required in qIVIVE for risk assessment purposes. Additionally, it shows that even for an assumed relatively data-rich chemical like CPF knowledge gaps are present, and these hampered the NAM-based risk assessment. Specifically, further research is needed on the selection and refinement of the endpoint *in vitro* and the construction of a quantitative link between the selected key event and the adverse outcome, including possible feedback/compensatory mechanisms. In conclusion, our work shows that NAM-based risk assessment requires a multidisciplinary approach and more attention should be paid to the uncertainties introduced in the different extrapolation steps to ultimately achieve regulatory implementation of NAMs.

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Conflict of interest

The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Data availability

Data described in the manuscript is available in the supplementary file[2].

Author contributions (All)

Conceptualization: E. Hessel, J. van Klaveren, M. Zeilmaker; *Methodology:* A. Zwartsen, S. Zhao, S. den Braver-Sewradj, V. de Leeuw, C. van Oostrom, V. de Bruijn, N. Kramer, E. Zwart, V. de Leeuw, B. Bokkers, G. Chen, M. Schepens; *Writing – original draft preparation:* A. Zwartsen, S. Zhao, J. Westerhout; *Writing – review and editing:* all authors.

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